



Kernovite

Geologists in the UK discover a new mineral - Kernovite, forming strikingly large emerald green crystals. <http://digit.in/feb21-01>



Mysterious Hovering Clouds

Operational Land Imager spots mysterious curious clouds atop the Transantarctic Mountains. <http://digit.in/feb21-02>

Eutrophication of a freshwater body, with excessive growth of algae and plankton

NUTRIENT POLLUTION

How an excess of nutrients is having an adverse effect on ecosystems around the world

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It may be difficult to think as nutrients that are something that causes pollution. Increased

agricultural activity combined with industrialisation has resulted in decades of humans pumping nitrogen and phosphorus into the environment. In the quantities that they have been injected, they are causing changes in grasslands all around the world. The most noticeable effect of nutrient pollution is overgrowth of algae, which can restrict the ability of organisms to find food, directly clog

the gills of fish, and make the water cloudy. The effects extend to the homes of humans. Shellfish that have consumed toxins released from some algae can then make those who eat them fall ill. Nutrient contamination in agricultural areas can pose a risk to infants.

Most of the approaches to tackle the menace of nutrient pollution is focussed around removing the algae. While there has been considerable effort to monitor the introduction of industrial pollutants into the environment, the effect of nutrient pollutants has not been considered to the same degree by policymakers. Nutrient pollution degrades water bodies, and with aggravated climate change, there is an increasing accumulation of these nutrients in the coastal regions, as all the pollutants are moved towards the edges of the continents by rivers and streams. It is not just the creatures living in the water bodies and their surroundings that are affected, everything on the planet is, including humans.

GRASSLAND FIRES

One of the main problems of nutrient pollution is additional plant growth, which is not necessarily a good thing for Mother Earth. Uncharacteristic vegetation may start growing on grasslands for example, allowing invasive species to spread, and force out native species. The additional plant growth can also increase the fire risk of the area. While agricultural activity introduces a large amount of nutrients into the surroundings, it is also replacing the wild herbivores with domestic herbivores. Zebras in Africa, guanacos in South America and Reindeer around the north pole are all being replaced by domestic grazing animals such as cattle, sheep and goats. While wild grazing animals do have a mitigating effect on the excess plant growth, they cannot contain the growth entirely on their own. As the grasslands become wetter, hotter and more fertile, the capacity for the wild grazers to reduce the vegetation decreases. All of these conditions are expected to aggravate